

Week 5

5.1 Applications of the Chain Rule

5.1.1 The Gradient Vector is Perpendicular to the Level Sets

Theorem 103. $f : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, Ω open, $c \in \mathbb{R}$, $S = f^{-1}(c)$ and $a \in S$. If f is differentiable at $\vec{a}$ with $\nabla f(\vec{a}) \neq 0$, then $\nabla f(\vec{a}) \perp S$ at $\vec{a}$.

Remark. By $\nabla f(\vec{a}) \perp S$ at $\vec{a}$, we mean that for any tangent vector $\vec{v} \in \mathbb{R}^n$ to S at $\vec{a}$, $\nabla f(\vec{a}) \perp \vec{v}$. Equivalently, for any curve $\vec{\gamma} : I \rightarrow S \subset \mathbb{R}^n$ with $\vec{\gamma}(0) = \vec{a}$, we have $\nabla f(\vec{a}) \perp \vec{\gamma}'(0)$.

Example 104. If $f(x, y) = x^2 + y^2$, $S = f^{-1}(25)$ is the circle centred at the origin of radius 5, and $a = (4, 3) \in S$, then $\nabla f(x, y) = (2x, 2y)$ and so $\nabla f(4, 3) = (8, 6)$, which is certainly normal to the circle at the point $(4, 3)$.

Example 105. Let $S = \{x^2 + 4y^2 + 9z^2 = 22\}$ be an ellipsoid centred at the origin. In order to find the tangent plane of S at the point $(3, 1, 1)$, we consider the function $f(x, y, z) = x^2 + 4y^2 + 9z^2$ so that $S = f^{-1}(22)$. Then $\nabla f(x, y, z) = (2x, 8y, 18z)$, or $\nabla f(3, 1, 1) = (6, 8, 18)$ which is perpendicular to S at $(3, 1, 1)$. That is, $(6, 8, 18)$ is a normal vector to the tangent plane, and hence the tangent plane has equation

$$((x, y, z) - (3, 1, 1)) \cdot (6, 8, 18) = 0,$$

which simplifies to $3x + 4y + 9z = 22$.

Proof. Let $\vec{\gamma} : (-\epsilon, \epsilon) \rightarrow S$ be any curve inside the level set S such that $\vec{\gamma}(0) = \vec{a}$. Then its derivative $\vec{\gamma}'(0)$ is a tangent vector to S at $\vec{a}$ (in fact, this is how we technically define tangent vectors to a manifold). Since the curve $\vec{\gamma}$ always lies within the level set S , we find that

$$f \circ \vec{\gamma}(t) = c, \quad \forall t \in (-\epsilon, \epsilon).$$

In particular, $(f \circ \vec{\gamma})'(t) = 0$ for any $t \in (-\epsilon, \epsilon)$. Alternatively, we can apply the chain rule to find that

$$(f \circ \vec{\gamma})'(t) = \nabla f(\vec{\gamma}(t)) \cdot \vec{\gamma}'(t), \quad \forall t \in (-\epsilon, \epsilon).$$

Setting $t = 0$, we find that $\nabla f(\vec{a}) \cdot \vec{\gamma}'(0) = 0$. Repeating for all possible curves $\vec{\gamma}$, we find that $\nabla f(\vec{a})$ is orthogonal to every tangent vector to S at $\vec{a}$. $\square$

Remark. By applying the chain rule to f composed instead with the straight line $\alpha(t) = a + tu$, for some unit vector $u \in \mathbb{R}^n$, we find that

$$D_u f(a) = (f \circ \alpha)'(0) = \nabla f(\alpha(0)) \cdot \alpha'(0) = \nabla f(a) \cdot u.$$

So the Chain Rule also gives an alternative proof of Lemma 92.

5.1.2 Implicit Differentiation

Consider the circle $\{x^2 + y^2 = 1\}$. How do we find $\frac{dy}{dx}$ at the point $(\frac{3}{5}, \frac{-4}{5})$?

Locally near $(\frac{3}{5}, \frac{-4}{5})$, we find that $y^2 = 1 - x^2$ and $y < 0$ implies $y = -\sqrt{1 - x^2}$ explicitly. So y is a function of x near $(\frac{3}{5}, \frac{-4}{5})$. We could then differentiate this formula to find what we wanted.

Alternatively, rather than finding an explicit representation of y as a function of x locally, we could utilise the Chain Rule. That is, for the equation $x^2 + y^2 = 1$, consider x as a variable and y as a function of x . Then, differentiating with respect to x , we have

$$2x + 2y \frac{\partial y}{\partial x} = 0.$$

Therefore, at the point $(\frac{3}{5}, \frac{-4}{5})$ we have

$$2(\frac{3}{5}) + 2(\frac{-4}{5}) \frac{\partial y}{\partial x} = 0,$$

which simplifies to $\frac{\partial y}{\partial x}|_{(\frac{3}{5}, \frac{-4}{5})} = \frac{3}{4}$.

Remark. Unlike at the point $(\frac{3}{5}, \frac{-4}{5})$, near the point $(-1, 0)$, y is not a function of x locally. In fact, when we put $(-1, 0)$ into our implicit formula for $\frac{\partial y}{\partial x}$ where we assumed y was locally a function of x , we end up with the nonsense $-2 + 0 = 0$.

Instead, x is locally a function of y at the point $(-1, 0)$, and so we have the implicit equation $2x \frac{\partial x}{\partial y} + 2y = 0$, which results in $\frac{\partial x}{\partial y}|_{(-1, 0)} = 0$.

Example 106. Let $S = \{x^3 + z^2 + ye^{xz} + z \cos y = 0\}$. Given that z can be expressed locally about $(0, 0, 0)$ as a function of x and y , find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ at $(0, 0, 0)$.

Differentiating our implicit formula for z as a function of x and y , we find

$$\begin{aligned} 3x^2 + 2z \frac{\partial z}{\partial x} + ye^{xz}(z + x \frac{\partial z}{\partial x}) + \frac{\partial z}{\partial x} \cos y &= 0 \\ 2z \frac{\partial z}{\partial y} + e^{xz} + ye^{xz}(x \frac{\partial z}{\partial y}) + \frac{\partial z}{\partial y} \cos y - z \sin y &= 0 \end{aligned}$$

Setting $x = y = z = 0$, we find $\frac{\partial z}{\partial x}|_{(0, 0, 0)} = 0$, and $\frac{\partial z}{\partial y}|_{(0, 0, 0)} = -1$.

5.2 Extremal Points

Definition 5.22. Let $f : A \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, $a \in A$.

(i) f is said to have a global maximum at $\vec{a}$ if

$$f(\vec{a}) \geq f(\vec{x}), \quad \forall \vec{x} \in A.$$

(ii) f is said to have a local maximum at $\vec{a}$ if

$$f(\vec{a}) \geq f(\vec{x}), \quad \text{for any } \vec{x} \in A \text{ near } \vec{a}.$$

More precisely, $\exists \delta > 0$ such that $f(\vec{a}) \geq f(\vec{x})$ for any $x \in A \cap B_\delta(\vec{a})$.

We make the exact same definitions for minimum values also.

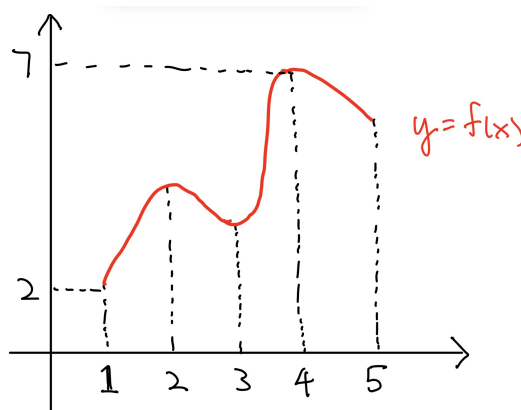


Figure 5.1: A function $f : [0, 5] \rightarrow \mathbb{R}$.

Example 107. Consider the function f in Figure 5.1. This function has a

- Global maximum at 4,
- Global minimum at 1,
- Local maximum at 2 and 4,
- Local minimum at 1, 3 and 5.

5.2.1 The Extreme Value Theorem

The following examples show that global extrema need not exist.

Example 108. $f(x) = e^x$ for $x \in \mathbb{R}$. Since $\lim_{x \rightarrow \infty} f(x) = \infty$, there is no global maximum. Also, as $\lim_{x \rightarrow -\infty} f(x) = 0$ and $f(x) > 0$ for any $x \in \mathbb{R}$, then there is also no global minimum.

Note: The domain $\mathbb{R}$ is not bounded.

Example 109. $f(x) = x$ for $x \in (-1, 1]$. There is a global maximum of 1, but f has no global minimum.

Note: The domain $(-1, 1]$ is not closed.

Example 110. $f : [-1, 1] \rightarrow \mathbb{R}$, defined piecewise by

$$f(x) = \begin{cases} 1 - x & : x \in (0, 1] \\ 0 & : x = 0 \\ -1 - x & : x \in [-1, 0) \end{cases}.$$

f gets arbitrarily close to ± 1 but never reaches them, and so f has neither a global maximum or a global minimum.

Note: The function f is not continuous.

The following theorem provides a sufficient condition, but not a necessary condition, for the existence of global extrema.

Theorem 111 (Extreme Value Theorem). *Let $A \subseteq \mathbb{R}^n$ be closed and bounded and $f : A \rightarrow \mathbb{R}$ be continuous. Then f has a global maximum and minimum.*

Remark. *Closed and bounded subsets $A \subseteq \mathbb{R}^n$ are also known as **compact** subsets.*

Although there is a more general definition of compact subsets in any topological space, in Euclidean space being compact is equivalent to being closed and bounded. For the interested reader, look up the Heine-Borel theorem.

Although we may use the Extreme Value Theorem (EVT) to verify the existence of global extrema, the question now turns to how one may locate them.

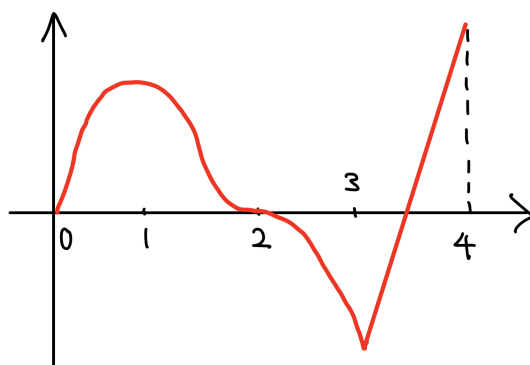


Figure 5.2: A function $g : [0, 4] \rightarrow \mathbb{R}$.

Example 112. Consider the function g in Figure 5.2. Firstly, since $A = [0, 4]$ is compact and f is continuous, by the EVT we know that f has a global maxima and minima.

Recall from single variable calculus, extrema can only occur at points x where

- (i) $f'(x) = 0$ (i.e at $x = 1, 2$)
- (ii) $f'(x)$ DNE (i.e at $x = 3$)

(iii) $x \in \partial A$ (i.e at $x = 0, 4$)

Points in cases (i) and (ii) are known as critical points and in case (iii) as boundary points.

Comparing the value of f at these five points, we find that f has a global max at 4 and a global min at 3.

We generalise the definition of a critical point to multi-variable functions.

Definition 5.23. $f : A \rightarrow \mathbb{R}$, $\vec{a} \in \text{Int}(A)$. Then a is called a critical point of f if either:

- $\nabla f(\vec{a})$ DNE;
- $\nabla f(\vec{a}) = 0$.

Theorem 113 (First derivative test). If $f : A \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ attains a local extremum at $\vec{a} \in \text{Int}(A)$, then a is a critical point of f .

Proof. Since $\vec{a} \in \text{Int}(A)$, we can find $\delta > 0$ such that $\vec{a} + t\vec{v} \in A$ for any $|t| < \delta$ and any unit vector $\vec{v} \in \mathbb{R}^n$.

Without loss of generality, we may assume that $\nabla f(\vec{a})$ exists. Then, for any $i \in \{1, \dots, n\}$ consider the function

$$g_i(t) = f(\vec{a} + t\vec{e}_i), \quad t \in (-\delta, \delta).$$

By the definition of g_i , we find that g_i has a local extremum at 0 and $g'_i(0) = \frac{\partial f}{\partial x_i}(\vec{a})$ exists. Therefore, by the first derivative test for single variable functions

$$\nabla f(\vec{a}) = (g'_1(0), \dots, g'_n(0)) = (0, \dots, 0) \in \mathbb{R}^n. \quad \square$$

5.2.2 Finding Extrema

We employ the following strategy for find extremal points:

1. Find critical points of our function in the interior $\text{Int}(A)$.
2. Find the max/min of our function on the boundary ∂A .
3. Compare these values.

Example 114. $f(x, y) = x^2 + 2y^2 - x + 3$, defined for $(x, y) \in A = \overline{B_1(0)}$.

Firstly, we note that A is compact and since f is a polynomial it is continuous, so by the EVT f has global extrema in A .

1. $\nabla f(x, y) = (2x - 1, 4y)$ exists everywhere in $\text{Int}(A) = B_1(0)$. Also, $\nabla f(x, y) = 0$ iff $(x, y) = (\frac{1}{2}, 0) \in B_1(0)$, with $f(\frac{1}{2}, 0) = \frac{11}{4}$.

2. We may parameterise ∂A by $\theta \mapsto (\cos \theta, \sin \theta)$, for $\theta \in [0, 2\pi]$. Plugging this into our function f we find that

$$\begin{aligned} f(\cos \theta, \sin \theta) &= \cos^2 \theta + 2 \sin^2 \theta - \cos \theta + 3 \\ &= \cos^2 \theta + 2(1 - \cos^2 \theta) - \cos \theta + 3 \\ &= 5 - \cos \theta - \cos^2 \theta \\ &= \frac{21}{4} - \left(\cos \theta + \frac{1}{2}\right)^2. \end{aligned}$$

From this, we can easily read off that the

- maximum value on ∂A is $\frac{21}{4}$ when $\cos \theta = -\frac{1}{2}$, i.e at the points $(-\frac{1}{2}, \pm \frac{\sqrt{3}}{2})$.
 - minimum value of ∂A is $\frac{21}{4} - \frac{3}{2} = 3$ when $\cos \theta = 1$, i.e at the point $(1, 0)$.
3. Comparing our critical points and our boundary points, we find that
- $f(\frac{1}{2}, 0) = \frac{11}{4}$ is the global min,
 - $f(-\frac{1}{2}, \pm \frac{\sqrt{3}}{2}) = \frac{21}{4}$ is the global max.

Example 115. $f(x, y) = \sqrt{x^2 + y^4} - y$, defined for $(x, y) \in A = \{-1 \leq x, y \leq 1\}$.

A compact and f continuous implies that f has global extrema by the EVT.

1. Consider f restricted to $\text{Int}(A) = \{-1 < x, y < 1\}$. At the origin we find that $\frac{\partial f}{\partial x}(0, 0) = \lim_{h \rightarrow 0} \frac{|h|^{-0}}{h}$ DNE. Away from the origin f is differentiable with

$$\nabla f(x, y) = \left(\frac{x}{\sqrt{x^2 + y^4}}, \frac{2y^3}{\sqrt{x^2 + y^4}} - 1 \right),$$

and so $\nabla f = 0$ iff $(x, y) = (0, \frac{1}{2})$. Thus, f has two critical points $(0, 0)$ and $(0, \frac{1}{2})$ in $\text{Int}(A)$, and $f(0, 0) = 0$, $f(0, \frac{1}{2}) = -\frac{1}{4}$.

2. We split ∂A into three components:

- If $y = 1$, then $f(x, 1) = \sqrt{x^2 + 1} - 1$, which implies $f(x, 1) \in [0, \sqrt{2} - 1]$.
- If $y = -1$, then $f(x, -1) = \sqrt{x^2 + 1} + 1$, which implies $f(x, -1) \in [2, \sqrt{2} + 1]$.
- If $x = \pm 1$, then $f(\pm 1, y) = \sqrt{1 + y^4} - y$, which implies $f(\pm 1, y) \in (0, \sqrt{2} + 1]$.

Therefore, on ∂A , the minimum value is 0 at $(0, 1)$, and the maximum value is $\sqrt{2} + 1$ at $(\pm 1, -1)$.

3. $f(0, \frac{1}{2}) = -\frac{1}{4}$ is a global minimum, and $f(\pm 1, -1) = 1 + \sqrt{2}$ is a global maximum.

5.2.3 Unbounded Regions

In practice, we may wish to find extremal points for functions defined on unbounded regions. Although we can't directly apply the EVT, in many cases we may still use it on a suitably chosen compact subset of our unbounded region.

Example 116. $f(x, y) = x^2 + y^2 - 4x + 6y + 7$ for all $(x, y) \in \mathbb{R}^2$. Firstly, we observe that $\lim_{(x,y) \rightarrow \infty} f(x, y) = \infty$, and so f has no global maximum on $\mathbb{R}^2$. As we shall see, f will have a global minimum.

The gradient $\nabla f(x, y) = (2x - 4, 2y + 6)$ exists on all of $\mathbb{R}^2$, with $\nabla f = 0$ iff $(x, y) = (2, -3)$. That is, there is only one critical point in $\mathbb{R}^2$, with $f(2, -3) = -6$.

We want to choose a $A \subseteq \mathbb{R}^2$ compact such that $(2, -3) \in \text{Int}(A)$, but also so that outside of $\text{Int}(A)$, f is larger than -6 .

Note that

$$\begin{aligned} f(x, y) &= x^2 + y^2 - 4x - 6y + 7 \\ &\geq x^2 + y^2 - 4(\sqrt{x^2 + y^2}) - 6(\sqrt{x^2 + y^2}) + 7 \\ &\geq (\sqrt{x^2 + y^2})(\sqrt{x^2 + y^2} - 10) + 7, \end{aligned}$$

and so if $\sqrt{x^2 + y^2} \geq 10$, $f(x, y) \geq 7$. Therefore, we chose $A = \overline{B_{10}(0)}$.

By the EVT, f restricted to A has a global minimum in A . Since there is only one critical point $(2, -3)$ in $\text{Int}(A)$, and $f \geq 7 > -6$ on ∂A , it follows that f has a minimum at $(2, -3)$ in A . Finally, since $f(x, y) \geq 7 > f(2, -3)$ for any $(x, y) \notin A$, f does indeed have a global minimum at $(2, -3)$.

Example 117. Suppose you are tasked with constructing a box (without a lid) that has a volume of $16m^3$. The material used to construct the base costs $\$2/m^2$, and the material for the sides costs $\$0.5/m^2$.

Q: What is the cheapest you can build such a box?

Suppose $x, y > 0$ denote the length (in m) of the sides of the base. It follows that the height of the box must then be $\frac{16}{xy}$. With these dimensions, the cost of the box is given by the function $C : \Omega = \{x, y > 0\} \rightarrow \mathbb{R}$,

$$\begin{aligned} C(x, y) &= 2xy + \frac{1}{2} \left(2 \frac{16}{xy} y + 2 \frac{16}{xy} x \right) \\ &= 2xy + \frac{16}{x} + \frac{16}{y}. \end{aligned}$$

We have formulated our problem as finding a global minimum of the function C on the unbounded domain Ω .

Contents

1. $\nabla C(x, y) = (2y - \frac{16}{x^2}, 2x - \frac{16}{y^2})$ exists everywhere in Ω , and $\nabla C = 0$ iff $(x, y) = (2, 2)$. That is, C has one critical point $(2, 2) \in \Omega$, with $C(2, 2) = 24$.
2. We now find a compact region $A \subseteq \Omega$ such that $(2, 2) \in \text{Int}(A)$ and $C > 24$ outside of $\text{Int}(A)$. We choose $A = [\frac{1}{10}, 1000] \times [\frac{1}{10}, 1000]$.

If $x \leq \frac{1}{10}$ or $y \leq \frac{1}{10}$, then $C(x, y) > \frac{16}{x} + \frac{16}{y} \geq 160$. Alternatively, if either $x \geq \frac{1}{10}$ and $y \geq 1000$, or $x \geq 1000$ and $y \geq \frac{1}{10}$, then $xy \geq 100$ and $C(x, y) > 2xy \geq 200$.

3. Since A is compact and C is continuous, by the EVT, C has a global minimum in A . Since $(2, 2) \in \text{Int}(A)$, and $C > 24 = C(2, 2)$ on ∂A , it follows that C has a minimum at $(2, 2)$ in A . Finally, since $C > 24$ at every point in $\Omega \setminus A$, C has a global minimum at $(2, 2)$ in Ω , and the cheapest price to construct the box is \$24.